

# TITLE PAGE

**Problem Statement ID: HW-BAD-2026**

**Problem Statement Title: AI-Powered Behavioral Anomaly  
Detection for Cybersecurity**

**Theme: AI/ML & Cybersecurity System Architecture**

**PS Category: Software**

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**GitHub Repository:**

**<https://github.com/SinghAnsh07/cybershield-AI>**

# IDEA TITLE: CyberShield AI

PROPOSED SOLUTION: Behavioral Anomaly Detection & Attack Classification

- \* Multi-Stage Anomaly Engine: Combines a Statistical Profiler (Mahalanobis Distance), a PyTorch Autoencoder, a 2-layer BiLSTM sequence detector with Focal Loss, and an XGBoost multi-class attack classifier.
- \* Solves Core Security Challenges: Handles class imbalance (<0.5% attacks via Focal Loss & SMOTE), concept drift (exponential age-decay sample weighting), and cold-start entities (cohort baseline blending).
- \* Plain-English Explainability: Integrates SHAP TreeExplainer to deliver human-readable alert reasons (e.g., 'Flagged due to impossible geo-velocity + novel device fingerprint') for fast SOC triage.
- \* Interactive SOC Dashboard: Streamlit analyst-facing application with KPI overview, ranked alert queues, and entity investigation timelines.

# TECHNICAL APPROACH & METHODOLOGY

## TECHNOLOGY STACK & PIPELINE FLOW

- \* Programming & Core Libraries: Python 3.12, NumPy, Pandas, Faker (Synthetic Generator), Scikit-Learn.
- \* Deep Learning & ML: PyTorch (Autoencoder & BiLSTM Detector), XGBoost (Multi-Class Classifier), Imbalanced-Learn (SMOTE).
- \* Explainability & UI: SHAP TreeExplainer, Streamlit Dashboard, Plotly Visualization Engine, FPDF2.
- \* End-to-End Execution Flow:
  1. Vectorized Log Ingestion -> 10-Field Telemetry Schema across Users, Service Accounts & Edge Devices.
  2. Feature Extraction Pipeline -> 26 behavioral features (geo-velocity, resource novelty, failed auth counts).
  3. Hybrid Profiling -> Autoencoder reconstruction MSE + Mahalanobis distance scoring.
  4. Sequence Anomaly Scoring -> 10-event window BiLSTM with Focal Loss.
  5. Attack Classification & Explanation -> XGBoost + SHAP natural language reason translation.
- \* Streaming Blueprint: Apache Kafka + Apache Flink stateful windowing in RocksDB & ONNX sub-50ms serving.

# FEASIBILITY, RESULTS & RISK MITIGATION

## EXPERIMENTAL RESULTS & OPERATIONAL VIABILITY

- \* Empirical Validation: Successfully evaluated on 212,475 access log events across 500 entities over 90 days.
- \* Low False Positive Rate: Achieved FPR = 0.91% at top 1% analyst alert budget (maintains strict SOC operational compliance).
- \* High Multi-Class Accuracy: Multi-Class Weighted F1 Score = 0.9117 across 7 injected attack types.
- \* Key Risk Mitigations:
  - Class Imbalance (<0.5% rare attacks) -> Focal Loss ( $\gamma=2.0$ ) + SMOTE minority oversampling.
  - Cold-Start Entities -> Cohort baseline blending smoothly transitions to individual profiles over 50 events.
  - Concept Drift -> Exponential age-decay weighting ( $0.95^t$ ) prioritizes recent behavioral trends.
  - High Event Throughput -> Kafka/Flink sub-50ms design with embedded ONNX runtime model scoring.

# SOLUTION ARTIFACTS & PROTOTYPE

## COMPLETE PRODUCTION DELIVERABLES

- \* GitHub Codebase: <https://github.com/SinghAnsh07/cybershield-AI> (Fully modular open-source implementation)
- \* End-to-End Pipeline: Single-command execution (``python pipeline.py``) for data generation, feature engineering, Autoencoder/BiLSTM/XGBoost training, evaluation, and report compilation.
- \* SOC Analyst Dashboard: Streamlit interactive UI (``streamlit run dashboard/app.py``) with risk metrics, alert queue, SHAP waterfall plots, and entity timelines.
- \* Production PDF Report: 5-page publication-quality technical document (``report.pdf``) with embedded ROC/PR curves.
- \* Trained Model Artifacts: Saved weights in ``data/models/`` (``baseline_profler.pt``, ``detection_model.pt``, ``anomaly_classifier.joblib``).

# RESEARCH & REFERENCES

## KEY REFERENCES & RESEARCH WORK

1. User and Entity Behavior Analytics (UEBA): IEEE Transactions on Information Forensics and Security on access log anomaly detection.
2. Sequence Anomaly Detection: PyTorch Bidirectional LSTM with Focal Loss for imbalanced temporal security events.
3. Model Explainability: Lundberg & Lee (2017), 'A Unified Approach to Interpreting Model Predictions' (SHAP TreeExplainer).
4. Real-Time Streaming Systems: Apache Flink Stateful Stream Processing & RocksDB state backend architecture.
5. Solution Source Code & Deliverables: <https://github.com/SinghAnsh07/cybershield-AI>